YASHASHWIN DORANALA

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SUMMARY

Mechanical design engineer with hands-on experience in Solid Edge, sheet-metal manufacturing drawings and assembly documentation, combined with software development and CAD automation skills. Built an in-house platform design tool that reduced modelling time from 16–24 hours to under 2 minutes. Interested in mechanical design, design automation and CAD customisation.

EXPERIENCE

GPI India — Associate Engineer II **Jun 2026 – Present**

- Modelled and drafted equipment across GPI's batter-line process, including powder hoppers, batter tanks and associated assemblies, translating lead engineers' specifications into detailed 3D models, fully defined 2D manufacturing drawings, sections, detail views and assembly documentation.
- Prepared manufacturing documentation for batter-line equipment, including special-component drawings, laser-cut DXF files created in ProgeCAD, and sheet-metal bend drawings in Solid Edge; uploaded the files to Ridder iQ ERP to support fabrication and assembly.
- Created and maintained Bills of Materials (BOMs) in Ridder iQ ERP and uploaded associated CAD and manufacturing files, including DXF files, bend drawings and special-component drawings, supporting accurate documentation and traceability.
- Followed lead engineer specifications and applicable drawing standards, including EN ISO 13920 for general tolerances on welded constructions, to prepare drawings clearly defined for manufacture and assembly.
- Developed an MCP integration for Claude with Solid Edge COM automation, enabling engineers to describe requirements in natural language and generate tailored 3D CAD models.
- Developed an automated platform designer that converts 2D layouts into complete platform assemblies—including floors, railings, stairs, openings, cross members and structural supports—in accordance with EN ISO 14122. Reduced modelling time from 16–24 hours to 1–2 minutes.

SAMSI Techno — Software Engineer **Jan 2026 – Jun 2026**

- Built a full-stack SaaS platform from scratch for idea management and product discovery, with configurable work items and forms, workflow-driven approvals, workspace collaboration, and AI-powered insights to summarise ideas and identify related items.
- Owned end-to-end development across frontend UI, backend APIs, database design, authentication, AI features and deployment.

PROJECTS

Coaxial Photo-Surveying Drone — Design, Manufacture and Test (Univ. of Liverpool) **Sep 2024 – Apr 2025**

- Created full 3D CAD assemblies and **GD&T manufacturing drawings** in Creo/Fusion from customer requirements and UK standards, and coordinated 3D printing, CNC and laser cutting.
- Wrote **MATLAB finite element models** (stiffness matrices, boundary conditions, nodal displacements and stresses) for the critical load-bearing parts.
- Integrated a Pixhawk flight controller with GPS, ESCs, BLDC motors and swashplate servos, and prepared the design reports and BOM.

FEA and Vibration Study of Aircraft Landing Gear **Sep 2024 – Nov 2024**

- Built custom MATLAB FEA code for **static, modal and transient analysis** and checked stresses, mode shapes and frequencies against **Abaqus**, which showed the critical members and resonance risks.

3D CFD Study of Concentrated Solar-Thermal Collectors (B.Tech thesis) **Jan 2024 – Jun 2024**

- Built an **ANSYS Fluent** model of a parabolic trough absorber with grooves and ribs (RNG k-ε, two-phase nanofluid, UDFs). The improved geometry with nanofluids achieved up to an **8.8× higher performance evaluation criterion**.

EDUCATION

Manipal Institute of Technology — B.Tech, Mechanical Engineering **2020 – 2024**

University of Liverpool, UK — MSc Aerospace Engineering coursework (studies interrupted) **2024 – 2025**

SKILLS

CAD: Solid Edge, ProgeCAD, SolidWorks, Creo, CATIA V5, Autodesk Fusion; engineering file and BOM handling in Ridder iQ ERP
Design: 3D modelling, 2D manufacturing drawings, sections and detail views, EN ISO 13920, sheet metal, special-part drawings, DXF and bend drawings, BOM management, assembly documentation, EN ISO 14122
Analysis: ANSYS Fluent, Abaqus, ANSYS Mechanical, MATLAB (FEA), SU2
Programming: C#, Python, TypeScript/JavaScript, Next.js, C, Solid Edge COM API automation
Languages: English, Hindi, Telugu, Spanish